

Customer Churn Forecasting System

Project Overview

This project implements a predictive analytics system designed to identify customers who are likely to leave a service (churn). The goal is to help businesses proactively detect high-risk customers and implement retention strategies before they leave.

Although the demonstration dataset uses financial transaction behavior (fraud dataset structure), the machine learning pipeline represents a complete churn prediction workflow including data preprocessing, feature engineering, model training, evaluation, and business insight generation.

The system uses multiple machine learning algorithms and compares their performance using industry-standard metrics.

Objectives

- Predict customers who are likely to churn
 - Analyze customer behavior patterns
 - Identify key factors influencing churn
 - Provide business insights to reduce churn rate
 - Enable proactive customer retention strategies
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Technologies Used

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Machine Learning Models:

- Logistic Regression
- Random Forest
- Gradient Boosting

Evaluation Metrics:

- Accuracy
 - Precision
 - Recall
 - F 1 Score
 - ROC-AUC
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Dataset Description

Dataset: Credit Card Transactions Dataset

The dataset contains transaction features used to train a classification model. Each record represents a financial transaction and is labeled as fraudulent or legitimate.

Key Columns:

- Time – Time elapsed between transactions
- Amount – Transaction amount
- V 1 –V 2 8 – PCA transformed numerical features
- Class – Target variable (0 = Normal, 1 = Fraud)

For the churn forecasting framework, the "Class" variable acts as the churn indicator.

Data Preprocessing

Data preprocessing ensures data quality before training machine learning models.

Steps performed:

- 1 . Dataset loading using Pandas
- 2 . Dataset exploration (shape, head, info, statistics)
- 3 . Missing value inspection
- 4 . Duplicate record removal

This step improves data reliability and reduces noise in the model.

Feature Engineering

Feature engineering improves predictive performance by creating meaningful variables.

New features created:

LogAmount

- Log transformation of transaction amount to normalize distribution

TransactionHour

- Extracted hour from transaction time

DayPeriod

- Categorized time into:
 - Night
 - Morning
 - Afternoon
 - Evening

Categorical encoding:

- One-Hot Encoding applied to DayPeriod variable

This helps models understand behavioral patterns related to transaction timing.

Exploratory Data Analysis (EDA)

EDA helps understand patterns and relationships in the dataset.

Analysis performed:

• Churn/Fraud distribution visualization • Transaction amount distribution • Fraud vs normal transaction comparison • Correlation heatmap

Insights from EDA:

- Fraud/churn cases are extremely rare
- Certain transaction patterns correlate with abnormal activity
- Transaction amount distribution is highly skewed

Handling Class Imbalance

Real-world churn or fraud datasets are highly imbalanced.

Problem:

Normal cases >> Churn cases

Solution:

Oversampling technique using sklearn resample.

Fraud samples are duplicated until the dataset becomes balanced.

Balanced dataset improves model learning and reduces bias toward the majority class.

Model Development

Three machine learning algorithms were implemented.

Logistic Regression

A statistical model used for binary classification.

Advantages:

- Interpretable
 - Fast training
 - Good baseline model
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Random Forest

An ensemble learning method using multiple decision trees.

Advantages:

- Handles nonlinear relationships
- Robust to noise
- Provides feature importance

Gradient Boosting

A boosting algorithm that sequentially improves prediction errors.

Advantages:

- High predictive accuracy
- Captures complex patterns

Model Evaluation

Models were evaluated using multiple performance metrics.

Accuracy

- Overall correct predictions

Precision

- Correct positive predictions

Recall

- Ability to detect churn cases

F 1 Score

- Balance between precision and recall

ROC-AUC

- Measures classification performance across thresholds

The model with the highest ROC-AUC score is selected as the best model.

Model Performance Comparison

All models are trained and evaluated on the test dataset.

A comparison table and visualization are generated showing:

- Accuracy
- Precision
- Recall
- F 1 Score
- ROC-AUC

This helps identify the best performing algorithm.

Best Model Selection

The best model is selected based on the highest ROC-AUC score.

ROC-AUC is preferred because it evaluates classification performance across multiple thresholds and is more reliable for imbalanced datasets.

Model Validation

Cross-validation is performed using 5-fold validation to ensure model stability and avoid overfitting.

This provides a more reliable estimate of model performance.

Model Interpretation

Feature importance analysis is performed when Random Forest is selected as the best model.

This helps identify the most influential predictors affecting churn.

Example influential factors:

- Transaction patterns
- Spending behavior
- Activity timing

Understanding these factors helps businesses design targeted retention strategies.

Risk Scoring System

Each transaction/customer is assigned a risk probability using the trained model.

Risk levels:

Low Risk (< 0.30)

Medium Risk ($0.30 - 0.70$)

High Risk (> 0.70)

Businesses can prioritize high-risk customers for retention campaigns.

Business Insights

The system generates actionable insights including:

- Fraud/churn risk segmentation
- Average fraudulent transaction amount
- Behavioral patterns of risky users

These insights help organizations:

- Reduce financial losses
 - Improve fraud detection
 - Enhance customer retention
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Business Value

This system helps companies:

- 1 . Detect risky behavior early
 - 2 . Identify high-risk customers
 - 3 . Launch targeted retention campaigns
 - 4 . Improve customer satisfaction
 - 5 . Reduce churn and financial losses
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Output Files Generated

Fraud_Risk_Predictions.csv

Contains:

- Prediction probabilities
- Risk segmentation

Model_Performance_Comparison.csv

Contains:

- Performance metrics for all models
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Future Improvements

Possible enhancements:

- Implement XGBoost or LightGBM
 - Apply SMOTE instead of simple oversampling
 - Deploy model using Flask or FastAPI
 - Build a real-time churn prediction dashboard
 - Integrate deep learning models
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Conclusion

This project demonstrates a complete machine learning pipeline for predicting customer churn using behavioral data.

By identifying high-risk users early, businesses can implement proactive retention strategies, reduce churn rates, and improve long-term profitability.

The system combines data analysis, feature engineering, and machine learning to provide both predictive power and actionable business insights.